Hisham Al-Zubair

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Education

Imperial College London — London, UK
MEng Electrical and Electronic Engineering — *Sept. 2026 – Present*

King Edward VI Camp Hill School For Boys — Birmingham, UK
*A*A*AA Maths, Further Maths, Physics, Computer Science* — *Sept. 2019 – Aug. 2026*

Experience

Energy Systems Design Work Experience — Oct. 2025 – Oct. 2025
Cundall — *Birmingham, UK*

- Proposed possible refurbishments to client's heating system, switching from gas to electric
- Designed a school to match the net zero goals set by local councils
- Explored laws regarding sustainable sourcing of energy

Coding Club Lead — Sep 2025 – May 2026
King Edward VI Camp Hill School For Boys — *Birmingham, UK*

- Led a weekly drop-in Python course for 23 students aged 11-14, designing tasks that progressed from basic syntax to small projects such as calculators
- Managed and trained a team of 4 assistants, delegating one-to-one support so that students of different abilities could progress at their own pace

Backend Team Leader - Engineering Society — Sep 2024 – May 2026
King Edward VI Camp Hill School for Boys — *Birmingham, UK*

- Proposed and led a team of 3 members to design the wiring architecture for a race car's battery connections, covering both power delivery to and from the battery
- Designed a latch mechanism on the car's back panels to give maintenance access to the battery at all times, engineered to withstand varied weather conditions and rough terrain.

Projects

SPI Master Controller | *SystemVerilog, Logisim, CocoTB(Python), GTKWave* — Oct 2026 – Present

- Designing a Serial Peripheral Interface (SPI) master controller in SystemVerilog, structured as modular RTL blocks with clearly defined interfaces between them
- Planned the full project by breaking it into sections, each with its own deadline, to keep development on schedule alongside a full university workload
- Implemented the first iteration of the clock divider and finite state machine (FSM), which generate the SPI clock and control the transfer sequence
- Building a verification flow using Cocotb to write Python testbenches, with GTKWave used to inspect waveforms and debug timing behaviour

32-bit Binary Multiplier | *SystemVerilog, Logisim, CocoTB(Python), GTKWave* — July 2026 – Aug 2026

- Designed a 32-bit arithmetic unit in SystemVerilog, combining a Kogge-Stone Adder and a Carry-Select Adder to balance speed against hardware cost
- Implemented multi-cycle control logic to complete each multiplication over several clock cycles, reusing hardware instead of building a fully combinational multiplier
- Built a Cocotb testbench that checked the design against a verified Python reference model, using randomised inputs across both large and small operand ranges
- Debugged timing and functional errors by inspecting waveforms in GTKWave, confirming correct results across 12,000 test cases

Key Skills

Technical Languages: Python, C++
Languages: English(Native), Arabic(Conversational), Mandarin(Beginner)
Developer Tools: VS Code, KiCad, Issie, Arduino IDE, LTSpice, GTKWave, Logisim
Libraries: pandas, NumPy, Matplotlib, PySpice, Cocotb
HDLs: Verilog, SystemVerilog
Awards: Simon Maddison Scholarship in Electrical and Electronic Engineering, UKESF Semiconductor Talent Award